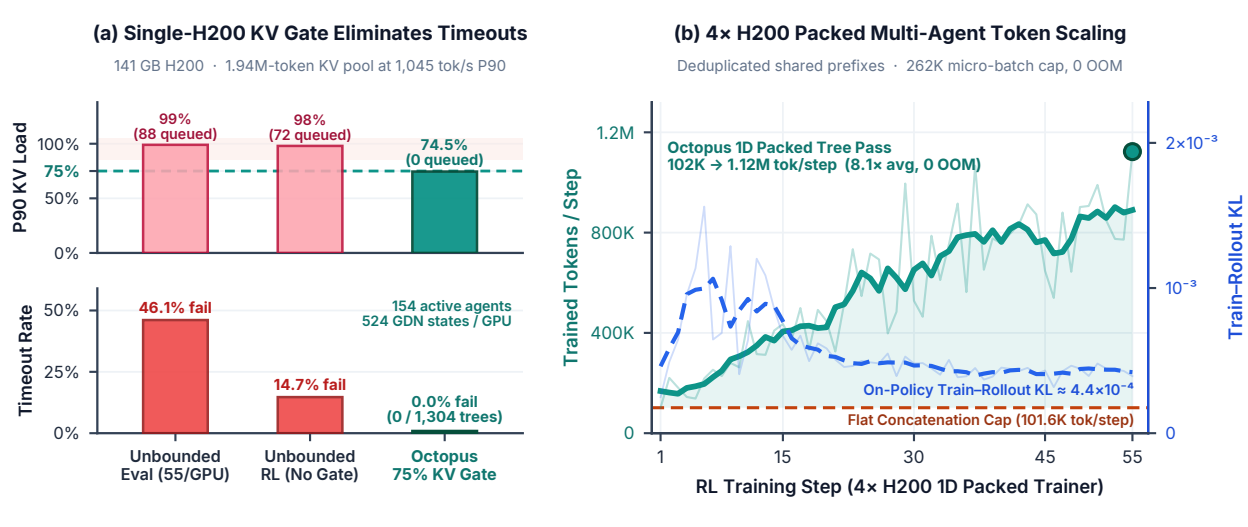
(a) Single-H200 KV Gate Eliminates Timeouts
141 GB H200 · 1.94M-token KV pool at 1,045 tok/s P90
P90 KV Load
100%
75%
50%
0%
99%
(88 queued)
98%
(72 queued)
74.5%
(0 queued)
Timeout Rate
50%
25%
0%
46.1% fail
14.7% fail
0.0% fail
(0 / 1,304 trees)
154 active agents
524 GDN states / GPU
Unbounded
Eval (55/GPU)
Unbounded
RL (No Gate)
Octopus
75% KV Gate
(b) 4× H200 Packed Multi-Agent Token Scaling
Deduplicated shared prefixes · 262K micro-batch cap, 0 OOM
Trained Tokens / Step
1.2M
800K
400K
0
Octopus 1D Packed Tree Pass
102K → 1.12M tok/step (8.1× avg, 0 OOM)
On-Policy Train–Rollout KL ≈ 4.4×10⁻⁴
Flat Concatenation Cap (101.6K tok/step)
2×10⁻³
10⁻³
0
Train–Rollout KL
1
15
30
45
55
RL Training Step (4× H200 1D Packed Trainer)